

Lecture Notes on
LINEAR ALGEBRA
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Linear Algebra For Econometrics

The main reason we need linear algebra in econometrics is that it provides a clean way to express and manipulate collections of variables and their relationships.

Matrices and Vectors

At its core, a *matrix* is just a rectangular array of numbers. In econometrics, we typically use matrices to represent our data, where rows correspond to observations and columns correspond to variables. For example, if we have N observations on K variables, we write this as an $N \times K$ matrix $\mathbf{X}$. This looks just like an Excel spreadsheet where each row is an observation and each column is a variable.

Each element of a matrix $\mathbf{A}$ can be referred to as $A_{i,j}$ where i is the row (observation) and j is the column (variable). A *column vector* is simply a $N \times 1$ matrix and typically is a 'variable'. When referring to a column of our matrix, we write $\mathbf{X}_{\cdot,j}$.

The transpose of a matrix, denoted $\mathbf{A}'$, flips rows and columns: the i -th column of $\mathbf{A}$ becomes the i -th row of $\mathbf{A}'$. In the transposed matrix, column j of the original matrix is the j -th row.

Matrix Operations

Matrix addition and subtraction work element-by-element, so dimensions must match. Scalar multi-

plication multiplies every element by a constant.

Matrix multiplication is more subtle. When we multiply an $r \times c$ matrix by a $c \times p$ matrix, the result is an $r \times p$ matrix. The key insight is that each element of the resulting matrix product is a *dot product* between a row of the first matrix and a column of the second. A dot product of two vectors is given by $x \cdot y = \sum_{i=1}^c x_i y_i$.

It's worth noting that matrix multiplication is associative and distributive, but *not* commutative: $\mathbf{AB} \neq \mathbf{BA}$.

Two special matrices are worth knowing. The **identity matrix** $\mathbb{I}_n$ acts like the number 1:

$$\mathbb{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

It has the property that $\mathbb{I}_n \mathbf{A} = \mathbf{A} \mathbb{I}_n = \mathbf{A}$.

The other is the **unit vector** $\iota = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$, which is

useful for summing elements.

Variance-Covariance Matrix

One particularly useful application is computing sample statistics. The sample variance of a variable

can be written as $\tilde{x}'\tilde{x}/(n-1)$ where $\tilde{x}$ is the de-meaned vector. Similarly, the variance-covariance matrix of multiple variables is

$$\frac{1}{n-1} \tilde{\mathbf{X}}' \tilde{\mathbf{X}}$$

where $\tilde{\mathbf{X}}$ is the data matrix with each column de-meaned. To see that this is indeed the variance-covariance matrix, note that the (i, j) -th element is:

$$\frac{1}{n-1} \sum_{k=1}^n (X_{k,i} - \bar{X}_i)(X_{k,j} - \bar{X}_j) = \widehat{\text{Cov}}(X_i, X_j).$$

The diagonal elements are variances (when $i = j$) and the off-diagonal elements are covariances between pairs of variables.

Matrix Transformations and Inverses

An $n \times n$ matrix times an $n \times 1$ vector can be thought of as a linear transformation: it can rotate, reflect, stretch, or shrink vectors in $\mathbb{R}^n$.

The *determinant* of a matrix tells us how much area changes under this transformation. If $\det(\mathbf{A}) = 0$, the transformation collapses dimensions and $\mathbf{A}$ has no inverse. When $\det(\mathbf{A}) \neq 0$, the inverse $\mathbf{A}^{-1}$ “undoes” the transformation: $\mathbf{A}^{-1}\mathbf{A} = \mathbb{I}_n$.

These concepts become essential when we derive the ordinary least squares estimator, which involves matrix inverses and products.